Loshitha Udana

Electrical and Electronic Engineering Undergraduate | AI, Computer Vision & Embedded Systems

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github.com/loshitha2002 LinkedIn Portfolio & evidence

Profile

Electrical and Electronic Engineering undergraduate with hands-on experience in artificial intelligence, machine learning, computer vision, embedded systems, and signal processing. Experienced in developing practical engineering solutions through research-oriented and collaborative projects. Strong analytical thinking, problem-solving, and programming skills, demonstrated through technical projects and competitive programming achievements. Passionate about applying emerging technologies to solve real-world challenges and continuously expanding technical expertise.

Education

University of Peradeniya – Faculty of Engineering Expected 2027
Undergraduate B.Sc. Engineering – Electrical and Electronic Engineering *Current GPA: 3.75*

St. Sylvester's College, Kandy G.C.E. Advanced Level
Physical Science Stream – Three A Grades (3As) *Z-Score: 2.0430*

Research Experience

Automated PCB Defect Detection System | *Computer Vision, PyTorch, CNNs, Edge AI* Ongoing

- Developing a learning-based pipeline to identify open-circuit, short-circuit and mouse-bite defects on bare printed circuit boards; exploring segmentation and robust image preprocessing.
- Investigating multi-scale feature fusion, model quantization and resource-constrained deployment; research **manuscript in preparation** (not yet published).
- Research progress / documentation

Selected Engineering Projects

InspectPin AI | *XIAO ESP32-S3 Sense, Python, React, Flutter* Ongoing

- Team-based portable electronics-inspection assistant integrating image/audio capture with an AI backend, web dashboard and mobile companion app; hardware acceptance testing remains in progress.
- GitHub repository

Autonomous Line-Following Robot | *Arduino, IR Sensors, Motor Control* Completed

- Developed a sensor-guided line-following robot; our team achieved **1st place among 60+ teams** in the GP118 competition.
- GitHub repository & project evidence

Elephant Detection Using Infrasound | *ESP32, FFT, k-NN, Python* Completed

- Developed a prototype analyzing low-frequency acoustic features to classify elephant presence, with a Python monitoring interface and ESP32 signal processing.
- GitHub repository

RFID Two-Factor Smart Door Lock | *ATmega328P, AVR Assembly, RFID* Completed

- Implemented PIN- and RFID-based access control in AVR assembly, using a keypad, MFRC522 module, finite-state-machine logic and a timed lockout after incorrect attempts.
- GitHub repository & technical documentation

Technical Skills

Programming: Python, C/C++, AVR Assembly, MATLAB; familiarity with Java.
AI / Computer Vision: PyTorch, CNNs, image segmentation, image preprocessing, feature extraction, model evaluation, quantization.
Embedded / Engineering: ESP32, Arduino/ATmega328P, sensors, digital signal processing, serial communication.
Tools: Git/GitHub, Google Colab, Kaggle, MATLAB/Simulink,Webots,PSCAD,Autocad,Raspberypi,Proteus, LaTeX/Overleaf.

Competitions and Achievements

VCoders 3.0 – Jaffna Coding *2024*
Final round – **5th overall** Result post

HaXtreme 4.0 *2025*
Final round – **7th overall** Result post

CodeArena 1.0 *2025*
Selected for physical/final round – Team BinaryBrains_UOP Selection list

SLIIT Xtreme *2025*
Qualified for national-level final round Result post

VCoders 4.0 – Jaffna Coding *2025*
Reached final round Certificate

Technovation *2024*
Selected for national university semifinal round Evidence on request

Programming Competition Participation

IEEEXtreme 19.0 *2025*
Sri Lanka rank: **71**; world rank: **894** (reported) Participation certificate

IEEEXtreme 18.0 *2024*
International programming competition participant Certificate

MoraXtreme 9.0 –
48th place (reported) Result post

MoraXtreme 10.0 *2025*
Algorithmic programming competition participant Certificate

Teaching Experience

G.C.E. A/L Combined Mathematics Tutor | *Sinhala Medium* Ongoing

– Teach Advanced Level Combined Mathematics, explaining challenging concepts, guiding problem-solving practice and preparing examination-oriented exercises.

Additional Information

Languages: Sinhala and English. **Interests:** AI/ML, computer vision, embedded intelligence and signal processing.
Verification: Linked certificates are evidence of participation; placement information is based on provided results and should be corroborated against official rankings where available.